(a) Shared memory of one CTA, block-major, NRC = 8

A: per-warp slice, x10[w]

w0	k0m0	k0m1	k1m0	k1m1
w1	k0m0	k0m1	k1m0	k1m1
w2	k0m0	k0m1	k1m0	k1m1
w3	k0m0	k0m1	k1m0	k1m1

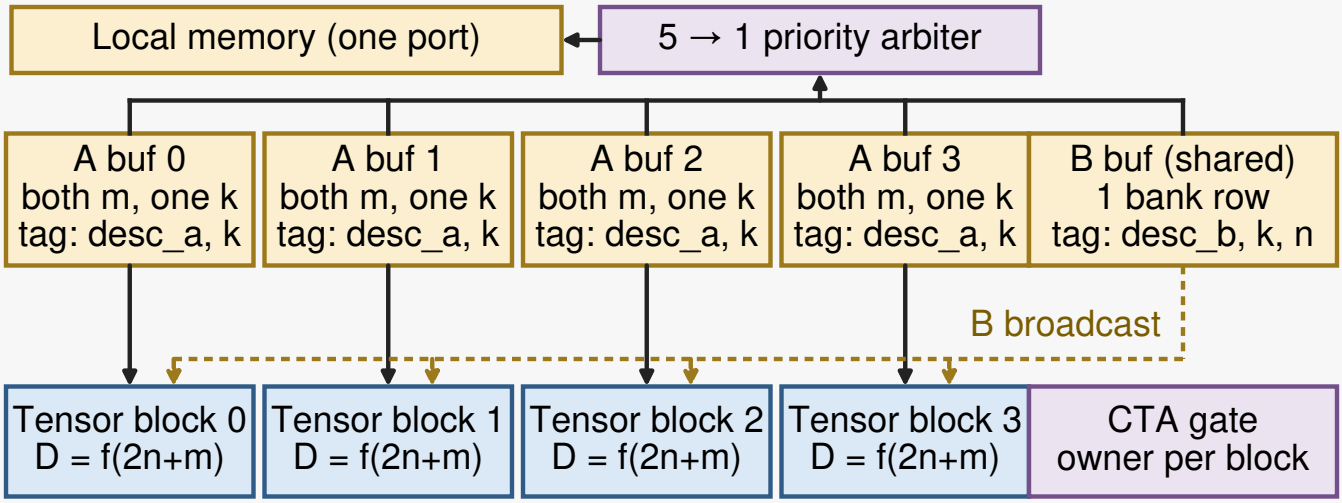
block = tcM × fedpK words

B: one tile for all warps, x11

k0	n0	n1	n2	n3
k1	n0	n1	n2	n3

bank row = 2 blocks (NT 8, 32),
1 block at NT 16 or with 2K FEDP
stride ≠ 0: row-major A, K-major B

(b) Engine: 4 + 1 tile buffers, one port, four tensor blocks



ready[b] = A buf[b] ∧ B buf ∧ key[b] ∧ ¬conflict[b]
key[b]: (desc_b, k, n, NRC) same as the leader block
conflict[b]: another block runs a different CTA
RS mode: A comes from f24–f27 and bypasses the A buffer

(c) One warp's micro-ops, SS mode, NRC = 8: k outer, n, m inner

one µop: a tcM × tcN output block over fedpK words of K

µop	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
k	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1
n	0	0	1	1	2	2	3	3	0	0	1	1	2	2	3	3
m	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1
acc	f0	f1	f2	f3	f4	f5	f6	f7	f0	f1	f2	f3	f4	f5	f6	f7
A buf	F	H	H	H	H	H	H	H	F	H	H	H	H	H	H	H
B buf	F	H	H	H	F	H	H	H	F	H	H	H	F	H	H	H
slot lock	fu_lock at µop 0 · held by this warp · fu_unlock at µop 15															
CTA owner	claimed at µop 0 · released at µop 15															

F: fetch from local memory, the µop stalls. H: resident hit.
A refills at each new k and at µop 0 of every instruction.